

Business Requirements Document

ARC — Agentic Receivables Recovery and Section 138 Legal Action Pipeline

Attribute	Value
Solution	ARC — Agentic Receivables Recovery and Section 138 Legal Action Pipeline
Client Context	PaintCo — decorative paints manufacturer, ~2,900 dealers, regional depots, TSIs
Primary Reference	Problem_Statement_1_Receivables_Recovery_Section138 (Organization-approved)
Version	2.0
Date	28-Aug-2026
Status	Draft for Business & Technical Review

Core principle: ARC is agentic in orchestration and recommendation — but deterministic in validation, authorization, money, dates, eligibility and legal persistence. The AI never computes an amount, date or eligibility verdict that appears in a legal notice. All such values come from the rules engine and metric contracts.

This BRD is a visual implementation guide. Read the flowcharts first.

1. Document Information

Item	Details
Document Owner	Business Process Owner — Receivables Recovery & Legal Action
Solution Owner	ARC Solution Team
Reviewers	Finance, Legal, Depot Operations, Field Sales, IT Architecture, Security
Source of Truth	Problem Statement 1 — Agentic Receivables Recovery and Section 138 Pipeline

1.1 Version History

Ver	Date	Change
1.0	28-Aug-2026	Initial BRD from Problem Statement
2.0	28-Aug-2026	Visual rewrite — 10 flowcharts, AI vs backend table, tools, TBC items; theory reduced

2. Executive Summary — What Is ARC?

ARC closes the gap between automated legal paper production and actual money recovery for PaintCo dealer receivables.

Question	Answer
What is ARC?	A human-gated, multi-day workflow system with eight AI agents (A1–A8) that reconcile receivables, prioritise recovery, recommend notices, orchestrate field visits, guard Section 138 eligibility, draft legal documents, and assemble court-ready case files.
What problem?	2,526 notices / ₹193.42 Cr chased in one month — but only 1 dealer visited, 0 rejections, unreconciled AR, manual S138 chain, no statutory clocks.
Two workflows	Workflow A — Monthly ODOS demand-notice cycle. Workflow B — Section 138 legal pipeline after 60 days non-payment and cheque bounce.
Who uses it?	Depot Manager, TSI, Depot Admin, ASM/HO, Legal, Advocate, Finance Controller
What AI does	Analyses, prioritises, recommends, drafts, orchestrates — via governed tools. Does not approve or compute legal amounts.
What humans do	Approve notices (G1), e-sign (G2), approve legal progression (G3), review case file (G4), confirm field PTP, correct cheque data

3. Problem Statement

ID	Problem	Live Impact
P1	Single hard trigger (ageing > 90 days AND balance > ₹5,000) — no recoverability ranking	2,526 notices vs capacity for ~few hundred visits/month
P2	Approval is timeout, not judgement — default Yes, 0 declines	Rubber-stamp gate
P3	Gross AR never reconciled — credits, rebates, returns, clearing cheques ignored	Wrongful notices; legal exposure
P4	Execution gap — 2,525 of 2,526 dealers never visited; ASM/HO remarks empty	Cannot tell "refused" from "nobody asked"
P5	S138 manual physical chain; statutory clocks in advocate's diary	Missed date extinguishes case
P6	Context fragmented across SAP, portal, PDFs, images, free-text	Inconsistent decisions
P7	No learning loop — same letter every month	Notices become noise

4. ARC Objective

#	Objective
1	Convert open-loop notice generation into closed-loop receivables recovery
2	Reconcile gross AR → net recoverable exposure before any action
3	Prioritise dealers by recoverability — not uniform treatment
4	Replace default-Yes with evidence-backed Issue / Hold / Reconcile
5	Instrument field follow-up — visit plans, Promise-to-Pay, broken-PTP chase
6	Guard S138 with eligibility checks and statutory limitation clocks
7	Assemble court-ready evidence bundles before filing
8	Measure which recovery lever actually works

5. Current Business Process

5.1 Process A — Monthly ODOS Demand-Notice Cycle (Today)

Step	Who	What Happens
1	Portal	1st — List dealers: ageing > 90 days AND balance > ₹5,000
2	Portal	1st — Bulk email to Depot Managers for dealer-by-dealer confirmation
3	Depot Manager	1st–7th — Approve (default Yes) or Decline with reason
4	HO Legal	After 7th — Generate notices with advocate signature; despatch
5	Field App	After 7th — Notice shown to covering TSI
6	TSI	Ongoing — Visit dealer; record feedback and recovery plan
7	System	Monthly — Cycle repeats for uncleared dealers

5.2 Process B — Section 138 Escalation (Today)

Step	Who	What Happens
1	Portal	List dealers: demand notice sent + no payment in 60 days
2	Depot Admin	Key cheque particulars; upload scan
3	Portal	Timeline freeze on 15th
4	Portal	Realised → close. Bounced → proceed
5	Depot Admin	Courier cheques to HO; key POD numbers
6	HO	Re-deposit cheques at single branch
7	Legal / PIS	Manually update bounce/realisation status
8	Legal	Draft Section 138 notice; upload per dealer
9	TSI / Court	Inform dealer; court summons; arrest warrant tracking (<i>manual, portal column exists</i>)

6. Proposed ARC Solution

ARC addresses nine sub-problems (SP1–SP9) through eight agents, two workflows, four human gates, and a deterministic rules engine — in Shadow, Assisted, or Live mode.

Sub-Problem	ARC Capability	Agent
SP1 Ledger reconciliation	Net recoverable exposure + line-item audit trail	A1
SP2 Entity resolution / Dealer 360	One dealer across SAP, portal, app, cheque, agreement	A1 + Knowledge
SP3 Recoverability scoring	Ranked worklist; top-decile in seconds	A2
SP4 Notice decisioning	Issue / Hold / Reconcile with cited evidence	A3
SP5 S138 eligibility	Validate debt, presentation, return reason before physical chain	A4
SP6 Limitation clock & drafting	Statutory dates; draft from graph facts only	A4 + A5
SP7 Field orchestration & PTP	Geo-clustered visits; voice → structured Promise-to-Pay	A6
SP8 Evidence assembly	Court-ready bundle + completeness score	A7

SP9 Supervision & learning	ASM/HO exceptions; lever effectiveness	A8
----------------------------	--	----

Sequencing: SP1–SP3 are upstream of everything. A1 must run in shadow mode before any agent influences a live notice.

7. Scope

7.1 In Scope

- Reconciliation, entity resolution, recoverability scoring, notice decisioning
- Four HITL gates (Depot Manager, Advocate, Legal ×2)
- S138 eligibility, limitation clocks, notice drafting, evidence assembly
- Field visit orchestration, PTP capture, broken-PTP escalation
- Shadow / Assisted / Live modes; durable multi-day workflows

7.2 Out of Scope

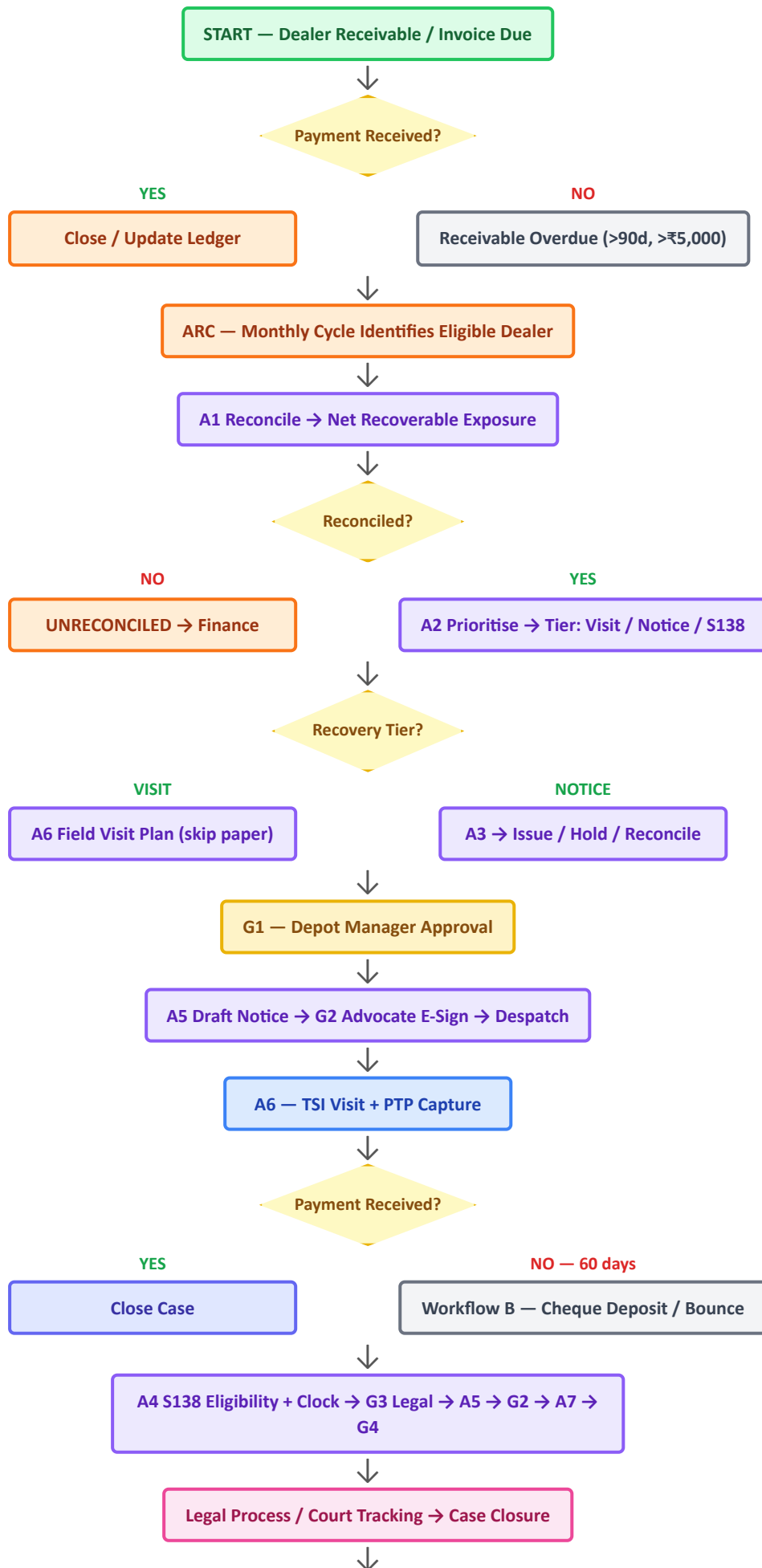
- Replacement of SAP / core ERP
- Court filing execution, hearings, arrest warrant automation (*portal tracks; no agent defined*)
- Payment collection / banking transactions
- Autonomous legal action without human gates

8. Actors / Users

Actor	Role in ARC
Depot Manager	Review evidence-backed recommendation; approve/decline/override (G1)
Depot Admin	Key cheque data; correct extraction mismatches; courier/POD
TSI	Execute visit tasks; confirm voice-captured PTP
ASM / HO Supervisor	Review exception queue (A8)
HO Legal / PIS	Eligibility review; legal progression (G3); case file review (G4)
Advocate	E-sign verified draft (G2); court proceedings
Finance Controller	Rule configuration; Reconcile-routed cases
Dealer (Debtor)	Receives notices; may pay, dispute or ignore — not a system user
AI Agents A1–A8	Reconcile, score, recommend, orchestrate, draft, assemble evidence
Backend / Rules Engine	Validation, amounts, dates, eligibility, persistence, authorization

9. ARC End-to-End Business Flow

FLOWCHART 1 — Complete ARC Flow (Workflow A + B entry)

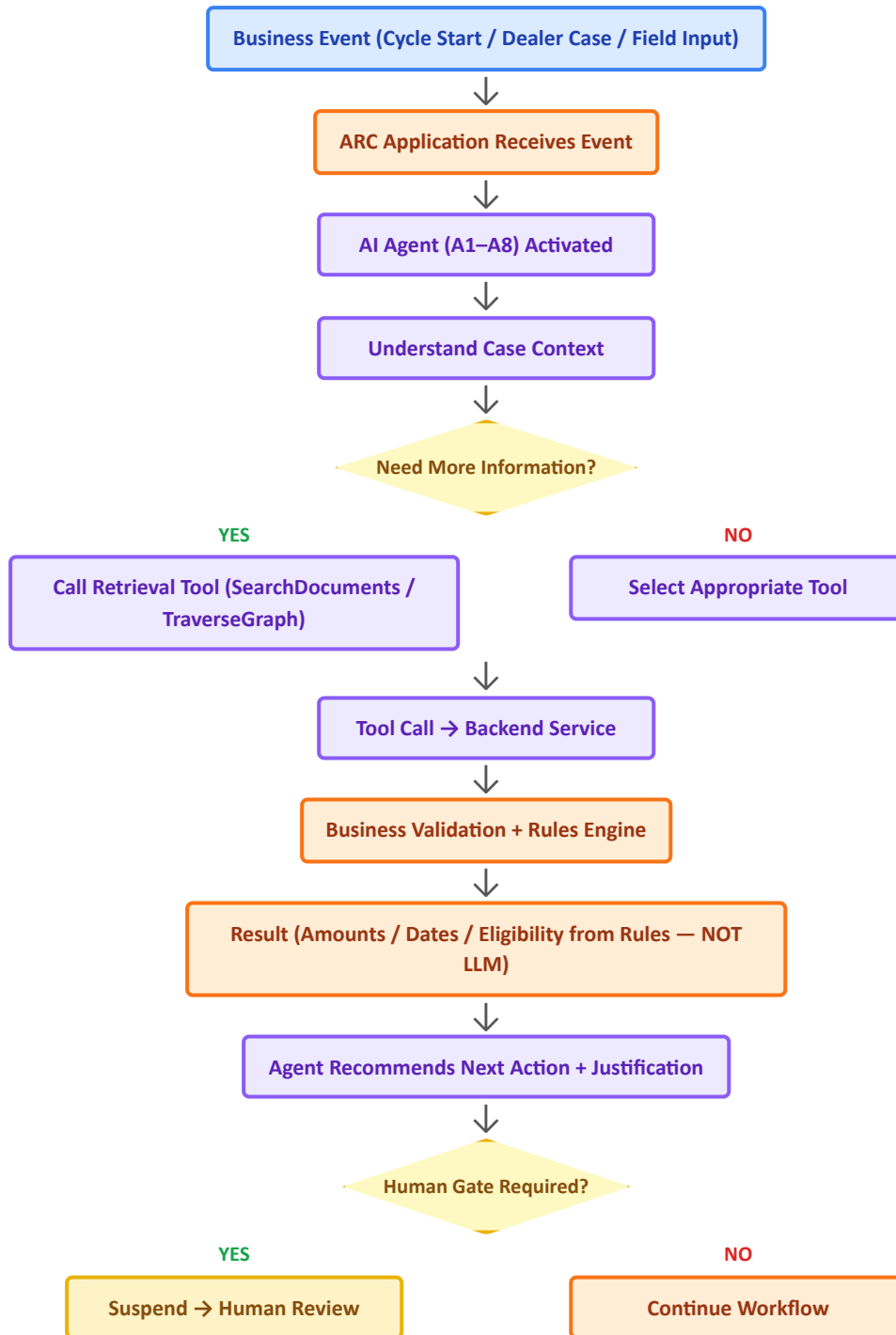


END

Step	Who	What Happens	System
1	ERP / Portal	Receivable ages past 90 days, balance > ₹5,000	SAP / Portal
2	Backend	Trigger monthly ODOS cycle	ARC Workflow A
3	A1	Compute net recoverable exposure	ComputeNetExposure tool
4	A2	Assign recovery tier	Rules + scoring
5	A3	Recommend Issue / Hold / Reconcile	SearchDocuments, TraverseGraph
6	Depot Manager	Approve or decline notice	Gate G1
7	A5 + Advocate	Draft and e-sign notice	Gates G2
8	TSI	Field visit and PTP	A6 + Field App
9	System	60 days no payment → S138 pipeline	Workflow B
10	Legal	Review eligibility, case file, filing readiness	Gates G3, G4

10. AI Agent Role

FLOWCHART 2 — What the AI Agent Does (Per Case Event)



10.1 Agent Decomposition (A1–A8)

Agent	Does	Output	Gate
A1 Reconciliation	Deterministic netting	net_recoverable_exposure + lineage	Auto
A2 Risk & Prioritisation	Scoring + graph features	Tier: Visit / Notice / Section138	Auto
A3 Notice Decisioning	Rules R1/R5 + evidence	Issue / Hold / Reconcile + citations	G1 Depot Manager
A4 Legal Eligibility & Clock	Rules R2/R3; LLM explains only	Eligible/blocked + file_by_date	G3 Legal

A5 Drafting	Template fill from graph facts; self-verify	Notice PDF	G2 Advocate
A6 Field Orchestration	Visit plans; voice → PTP	Visit tasks, PTP records	TSI confirm
A7 Evidence & Case File	Bundle assembly + completeness	Court-ready bundle	G4 Legal
A8 Supervisory Insight	Exception queue; lever analytics	ASM/HO exceptions	Auto

10.2 Information the AI Agent Needs

Data	Source	Used By
Gross AR, credit notes, rebates, returns, clearing cheques	SAP / Azure SQL	A1
Dealer identity across systems	SAP, Portal, Field App, Agreement	A1, A2
Payment history, ageing, dispute state	Ledger + Portal	A2, A3
Dealer agreement, prior notices, delivery proof	Blob / Vector store	A3, A7
Security cheque image, return memo	Blob + Document Intelligence	A4, A5, A7
TSI field remarks, visit history	Field App / Cosmos DB	A2, A6, A8
Statutory clock anchors	Rules engine (R3)	A4

10.3 Tools the AI Agent Calls

Tool	Purpose	Used By
ComputeNetExposure	Net recoverable exposure with deduction breakdown and source lineage. <i>Only authoritative source of claim amount.</i>	A1, others needing exposure
CheckSection138Eligibility	S138 eligibility per rules R2	A4
GetLimitationClock	Statutory dates and days remaining	A4
SearchDocuments	Hybrid search over policies, notices, agreements	A3, A8
TraverseGraph	Knowledge graph traversal (Dealer 360)	A2, A3, A4, A7

No SubmitNotice tool. Notices are emitted by the workflow after human gate approval — not by the model calling a function.

10.4 AI Agent vs Backend vs Human

Responsibility	AI Agent	Backend / System	Human
Understand case / context	✓		
Analyse receivable information	✓		
Prioritise / recommend action	✓		
Generate drafts & justifications	✓		
Select & call governed tools	✓		
Compute amounts / dates / eligibility		✓	
Apply business rules R1–R6		✓	
Database update / persistence		✓	
Authorization / RBAC		✓	
Notice approval (G1)			✓ Depot Manager

E-signature (G2)			✓ Advocate
Legal progression (G3) & case file (G4)			✓ Legal
Confirm voice PTP			✓ TSI
Final legal decision / court filing			✓ Legal / Advocate

AI MAY:

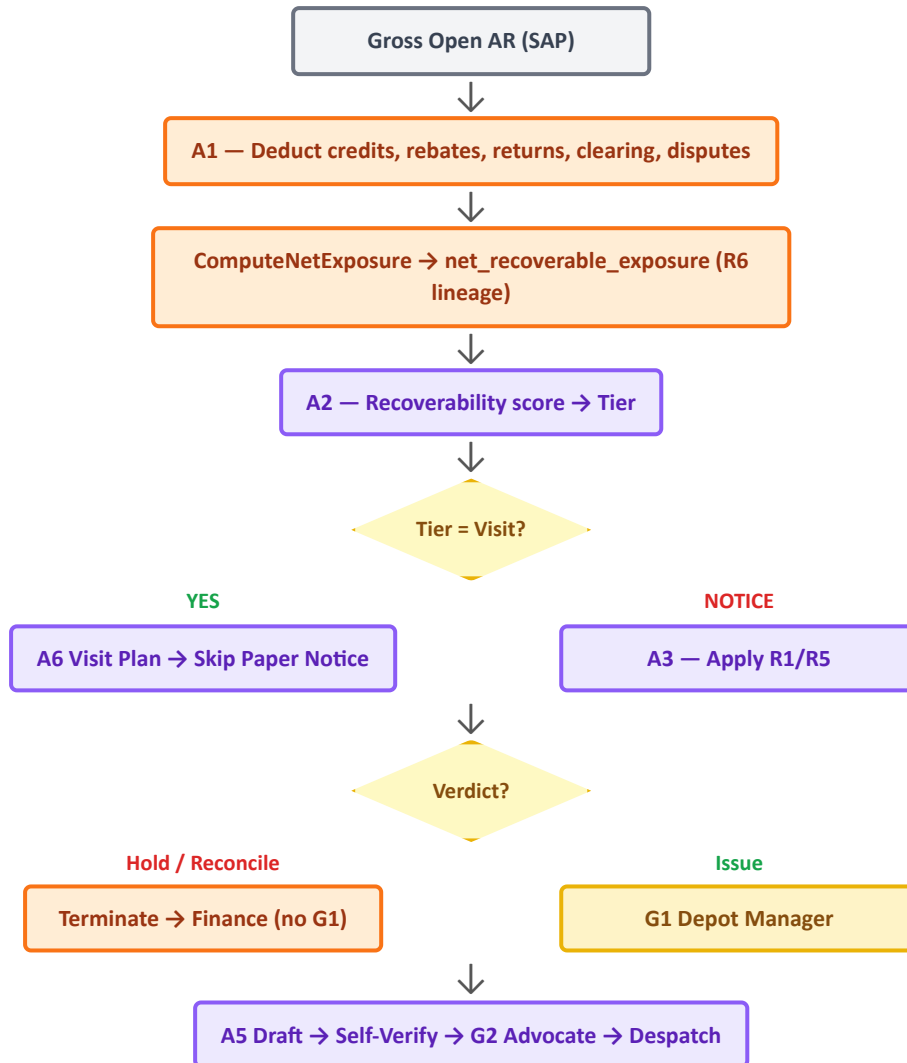
- Recommend Issue/Hold/Reconcile
- Draft notices from graph facts
- Plan visits; parse PTP
- Assemble evidence bundles
- Summarise recovery history

AI MUST NOT:

- Approve legal action
- Compute legal amounts/dates/eligibility
- Bypass rules or gates
- Modify financial records
- Call SubmitNotice

11. Receivables Recovery Flow

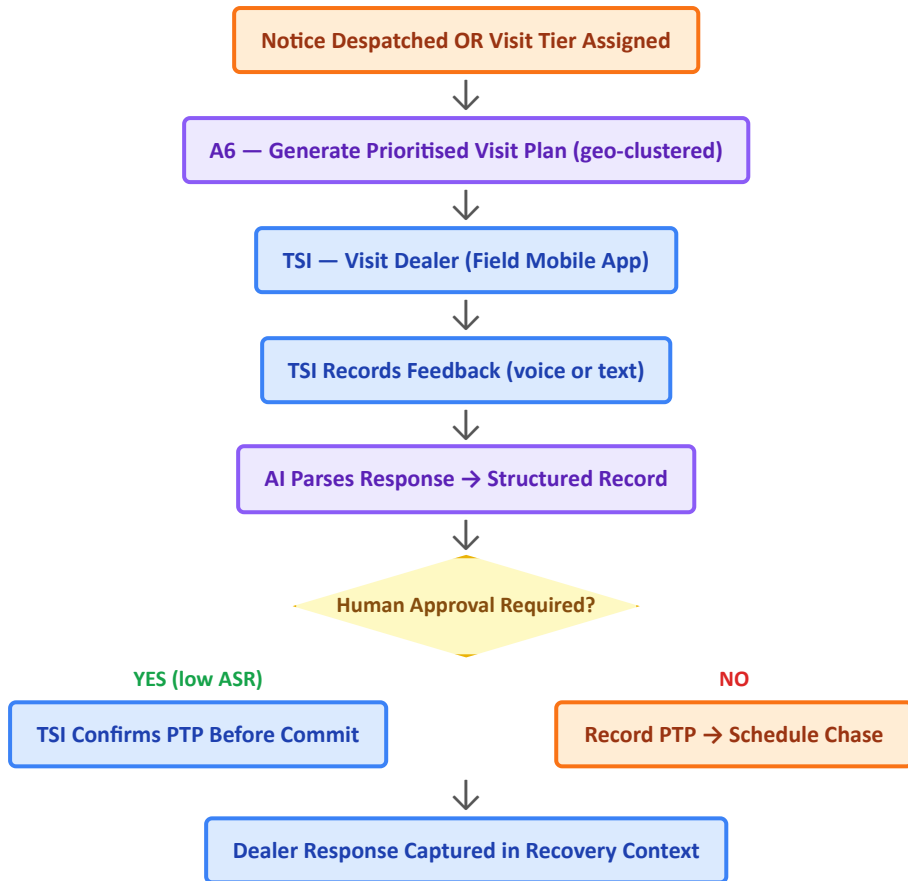
FLOWCHART 3 — Receivables Recovery (Workflow A — ODOS Cycle)



Metric: $\text{net_recoverable_exposure} = \text{gross_open_AR} - \text{unapplied_credit_notes} - \text{accrued_scheme_rebates} - \text{goods_return_in_transit} - \text{cheques_in_clearing} - \text{disputed_under_review}$

12. Customer / Dealer Follow-up Flow

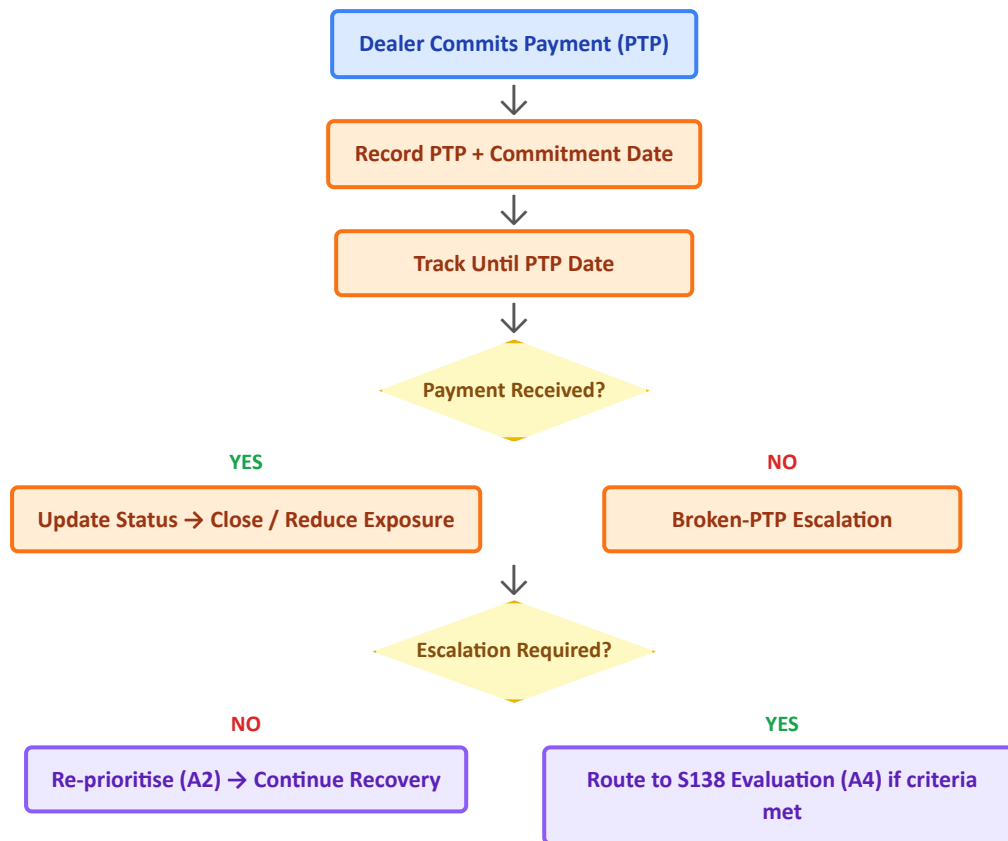
FLOWCHART 4 — Field Communication & Dealer Contact



Step	Who	What	Channel
1	HO Legal	Demand notice despatched to dealer	Physical / registered post
2	TSI	Prioritised visit task assigned	Field mobile app
3	TSI	Visit dealer; record feedback	Field app + voice
4	A6	Parse vernacular voice → PTP	Azure AI Speech + Agent
5	TSI	Confirm if ASR confidence low	Field app

13. Payment Commitment / Payment Tracking Flow

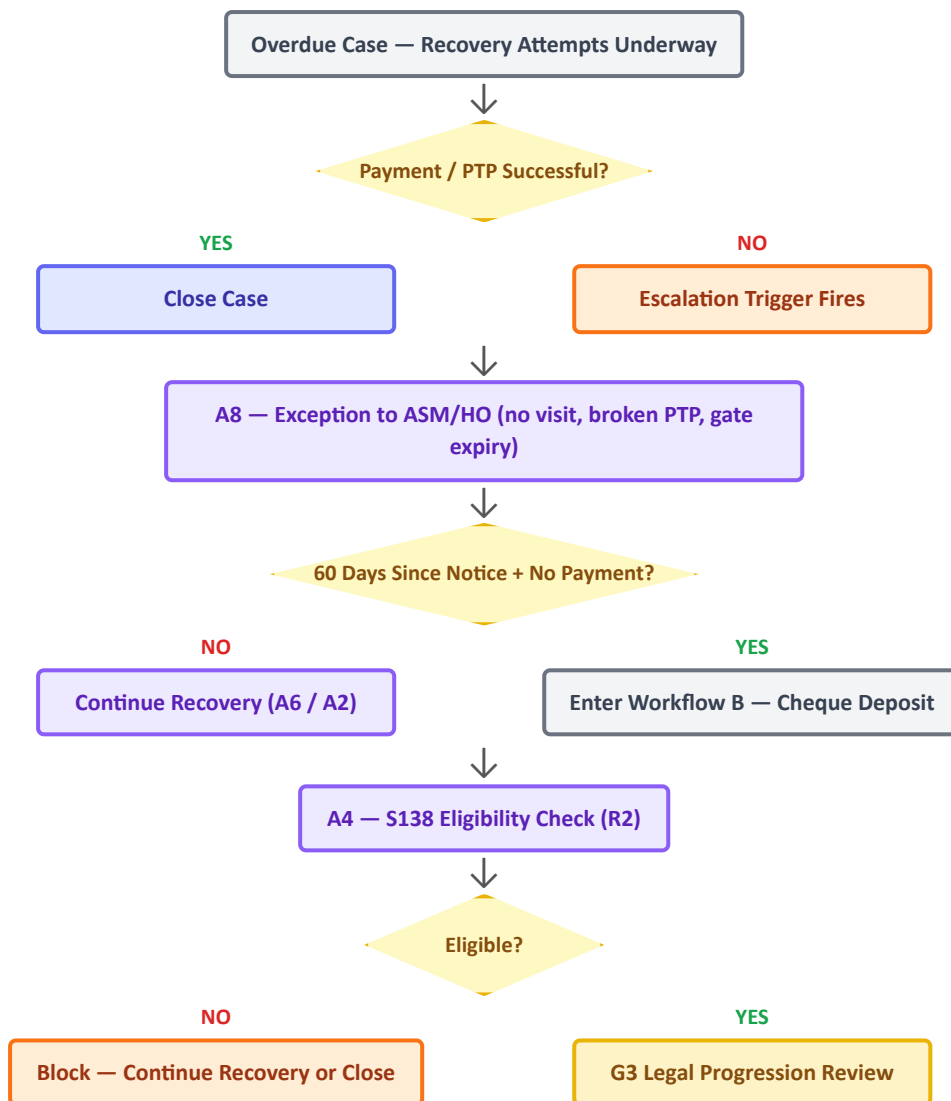
FLOWCHART 5 — Promise-to-Pay (PTP) Tracking



Step	Who	What
1	TSI / A6	Capture PTP date and amount (voice → structured)
2	Backend	Schedule chase date; block notice if active PTP within grace (R1c)
3	ERP	Payment receipt updates ledger
4	A6 → A2	Broken PTP triggers re-prioritisation or escalation

14. Escalation Flow

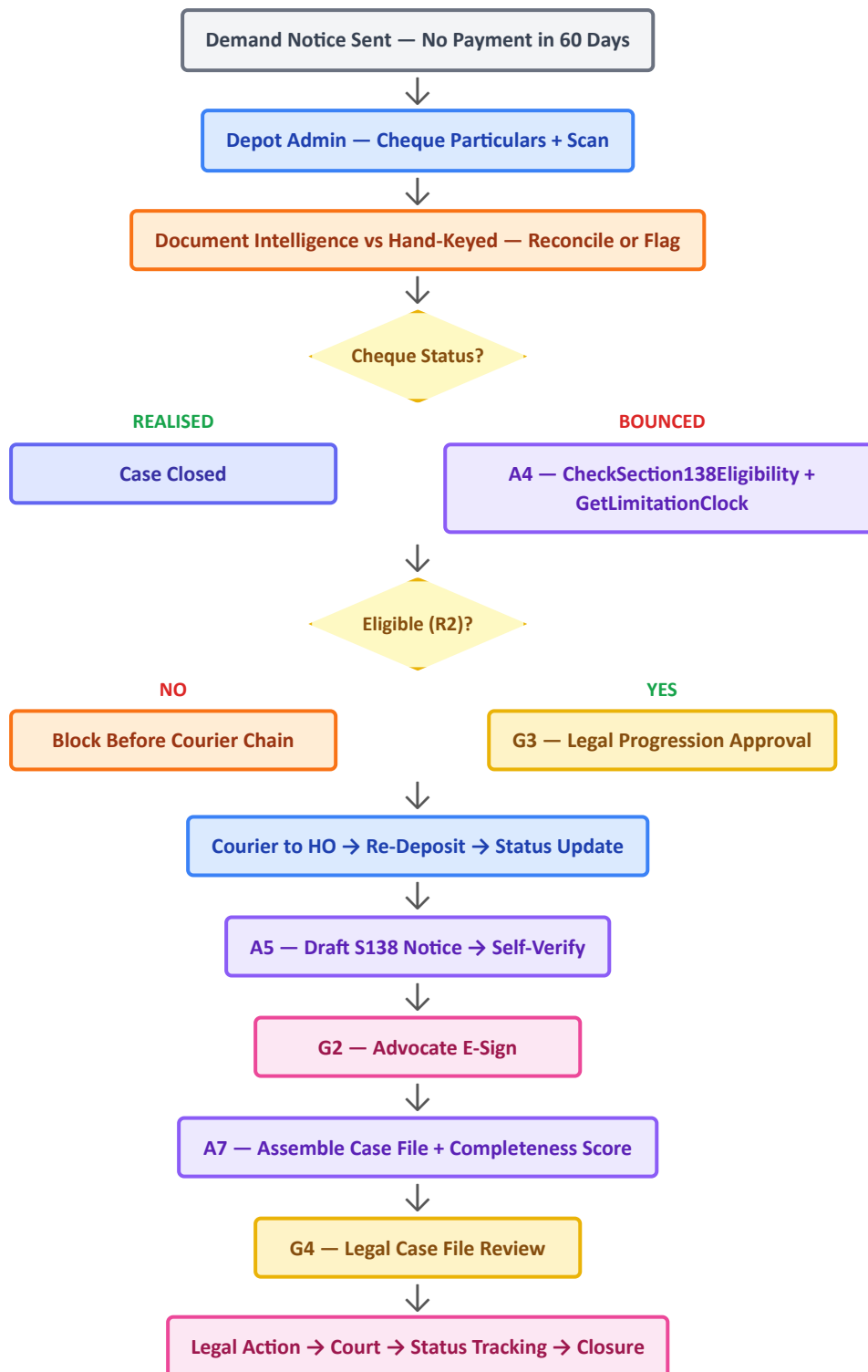
FLOWCHART 6 — Escalation Path



Trigger	Action	Owner
R1b — credit ratio > 40%	Reconcile → Finance; no notice	Rules Engine
Broken PTP	Re-prioritise or escalate	A6 → A2
No TSI visit on prioritised dealer	Exception to ASM/HO	A8
Limitation T-10 / T-5 / T-2	Alert named legal owner	A4 / R3
Clock expired	Case blocked — Expired	Rules Engine
Notice + 60 days no payment	Workflow B entry	Portal rule

15. Section 138 Eligibility / Legal Action Flow

FLOWCHART 7 — Section 138 Process (Workflow B)



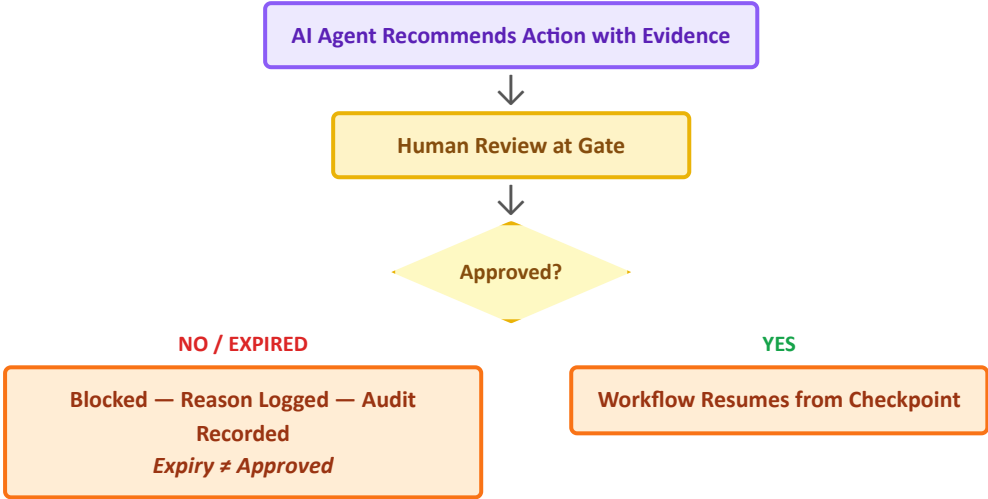
Step	Who	What	After Legal Action Starts
1	A4	Compute notice-by, cure-window, file-by dates	Alerts at T-10/T-5/T-2
2	Legal (G3)	Approve progression before physical chain	Courier / re-deposit proceeds
3	A5 + Advocate	Draft and e-sign S138 notice	Notice served to dealer
4	A7 + Legal (G4)	Court-ready bundle reviewed	Ready for filing

5	Legal / TSI	Court summons, warrant tracking	Manual — portal column; no agent defined (TBC)
---	-------------	---------------------------------	--

R2 eligibility: Enforceable debt in ledger + valid presentation + return-memo reason code in qualifying set. Missed statutory date **extinguishes** the case.

16. Human Approval / Human-in-the-Loop

FLOWCHART 8 — Human-in-the-Loop (All Gates)

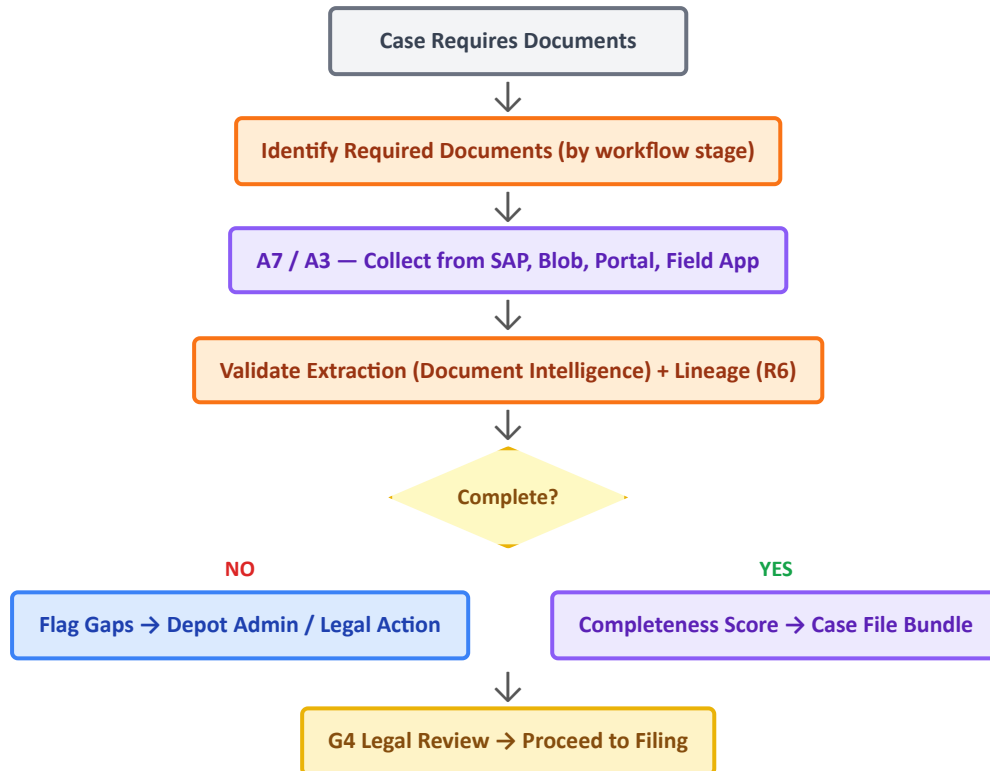


Gate	Name	Approver	When	Expiry
G1	gate_depot_manager	Depot Manager	After A3 recommends Issue only	Declined — NOT approved
G2	gate_advocate_signature	Advocate	After A5 self-verification passes	Case held — no dispatch
G3	gate_legal_progression	Legal Team	After A4 in Workflow B	Case blocked
G4	gate_legal_case_file_review	Legal Team	After A7 assembly	Case held — gaps flagged

Also human: TSI confirms low-confidence PTP; Depot Admin corrects cheque extraction mismatches. **R4:** Recommending agent cannot approve.

17. Documents and Evidence

FLOWCHART 9 — Document / Evidence Assembly



Document	Used In	Source
Ledger extract / invoices	Reconciliation, notice amount	SAP
Credit notes, rebates, returns	Net exposure (A1)	SAP
Dealer agreement	Entity resolution, enforceability	Blob
Security cheque image	S138 eligibility	Blob + Doc Intelligence
Cheque return memo	R2, limitation clock anchor	Blob + Doc Intelligence
Demand notice + delivery proof	S138 prerequisite	Blob (WORM, 8-year)
Courier POD	Physical chain evidence	Portal / Blob
Field visit + PTP	Recovery tracking	Field app
S138 notice draft	Legal action	A5 → G2
Court-ready bundle	Filing	A7 + completeness score

18. Business Rules

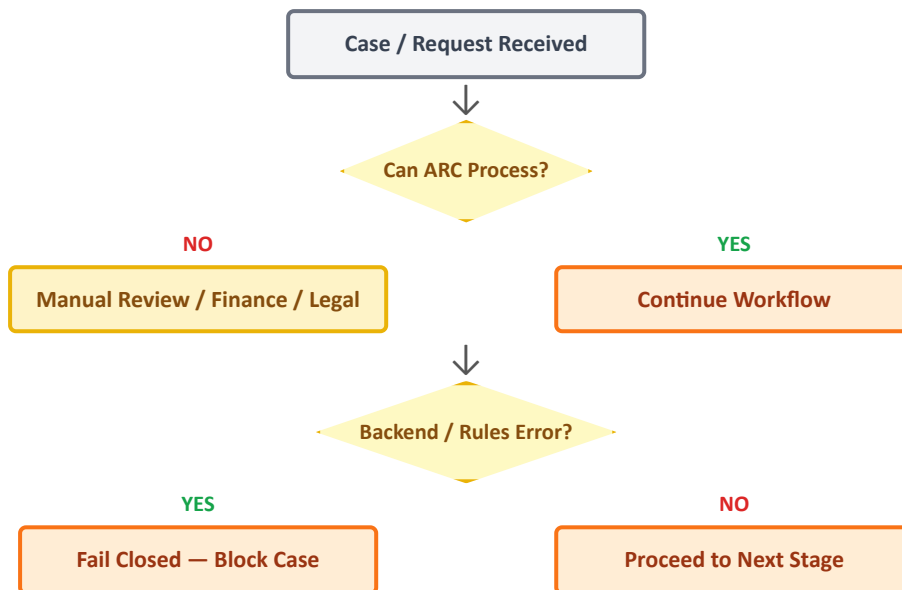
Rule	Business Rule	Owner
R1a	Hold notice if position unreconciled	Rules Engine
R1b	Hold if unapplied_credits / claim_amount > 40%	Rules Engine
R1c	Hold if Dispute UNDER_REVIEW or active PTP within grace	Rules Engine
R2	s138_eligible: enforceable debt + valid presentation + qualifying return reason	Rules Engine (A4)

R3	Statutory clock; alert T-10, T-5, T-2 to named owner	Limitation Clock Service
R4	Segregation of duties — recommender ≠ approver	Workflow design
R5	Block all action if insolvency moratorium	Rules Engine
R6	No notice amount without lineage to source rows	Metric Contract (A1)

Rules are versioned code owned by Legal/Finance — not prompt text.

19. Exception / Failure Scenarios

FLOWCHART 10 — Exception Handling



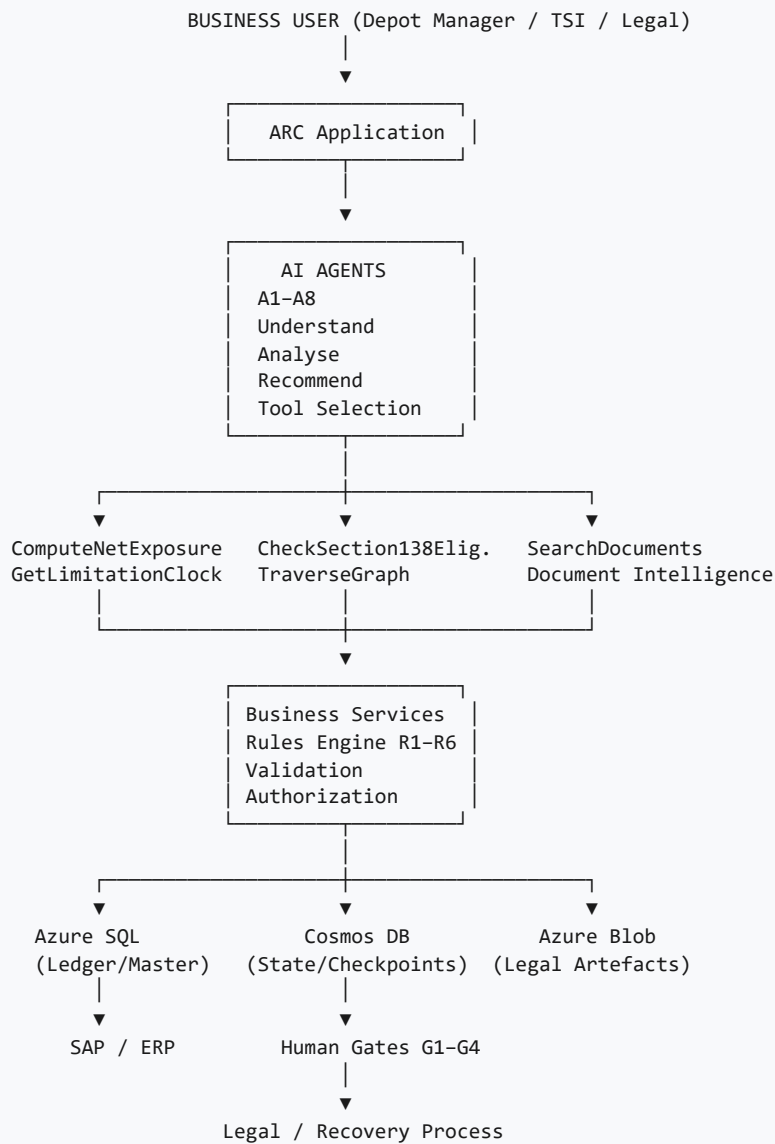
Scenario	Behaviour
A1 cannot reconcile	UNRECONCILED → Finance; never assume owed
Rules engine / clock unavailable	Fail closed — no case proceeds
A5 self-verification mismatch	Block draft — no best-effort notice
Low ASR on voice PTP	TSI confirmation; hard floor → discard
Duplicate dealer SAP codes	Entity resolution merge before exposure
Credit note after approval, before despatch	Re-validate; block if exposure changed
Cheque MICR contradicts portal entry	Flag Depot Admin; block progression
Non-qualifying return reason	R2 blocks before courier
Insolvency moratorium mid-cycle	R5 blocks every action
Depot Manager never responds	Gate expires — NOT approved
Checkpoint restore on different instance	Resume from persisted state
Overlapping cycles same dealer	Idempotency key prevents double-issue

20. Security and Audit

Area	Requirement
Authentication	Microsoft Entra ID; managed identity end-to-end
Authorization	RBAC server-side — TSI cannot read other regions
Segregation	R4 — recommender ≠ approver
PII	Redaction before model calls
Secrets	Azure Key Vault — no connection strings in config

Legal artefacts	WORM Blob, 8-year retention
Audit	Every gate decision: actor, role, verdict, reason, timestamp, correlation ID
Traceability	Every amount traceable to metric contract + source rows (R6)
Observability	OpenTelemetry: arc.cycle → arc.dealer → arc.executor → arc.gate
Shadow mode	Full cycle computable with zero external side effects

21. High-Level Architecture



21.1 Workflow Topology

Workflow A (ODOS): A1 → A2 → A3 <G1> → A5 <G2> → A6
| Visit tier → A6 [skip paper]
| Reconcile/Hold → TERMINATE → Finance

Workflow B (S138): A1 → A4 <G3> → A5 <G2> → A7 <G4>

22. Acceptance Criteria

#	Criterion	Target
1	Both workflows with 4 HITL gates as RequestPort suspensions	Required
2	Workflow survives restart; resumes from checkpoint	Required
3	Amounts/dates/eligibility traceable to rules + source rows (R6)	100%
4	No LLM output as final authorization	Zero tolerance

5	Rules R1–R6 unit-tested, versioned	Required
6	Statutory dates by rules engine; T-10/T-5/T-2 alerts	Zero misses
7	Wrongful-notice rate	< 1%
8	RBAC enforced server-side	Required
9	Full cycle in shadow mode — issue nothing	Required

22.1 Demo Scenarios (S1–S9)

ID	Scenario	Expected
S1	Clean overdue, no credits	Notice approved → drafted → visit task
S2	77% gross AR is credit note	R1b blocks → Reconcile → Finance
S3	Bounced security cheque	Full path to S138 case file
S4	Non-qualifying bounce reason	R2 blocks before courier
S5	3 days left on filing clock	T-2 alert to named owner
S6	Dealer under moratorium	R5 blocks all action
S7	Voice PTP at 0.72 confidence	TSI confirmation required
S8	Gate approval 6 days after pause	Resume from checkpoint
S9	Duplicate dealer identity	Entity merge before exposure

23. Summary

ARC in one page:

PaintCo has ~2,900 dealers. When receivables go overdue (>90 days, >₹5,000), ARC runs two workflows: **ODOS demand-notice recovery** and **Section 138 legal escalation** (after 60 days non-payment + cheque bounce). Eight AI agents reconcile AR, prioritise dealers, recommend notices, orchestrate field visits, guard legal eligibility, draft documents, and assemble evidence — always through governed tools and human gates. The backend rules engine owns all amounts, dates, and eligibility. Humans approve at four gates. Read Sections 9–19 flowcharts to understand the complete process.

23.1 To Be Confirmed (Not Invented in This BRD)

Item	Status
Exact statutory periods (notice/cure/filing windows)	Confirm with Legal counsel — versioned config, not prompts
Clock anchor for dishonour (memo received vs issue date)	Policy decision required
Second bounce — which dishonour anchors clock	Policy decision required
15th timeline freeze mapping to agent workflow	Not mapped in Problem Statement topology
Arrest warrant escalation automation	Portal column exists; no agent/workflow defined
Hold vs Reconcile handling in A3 termination	Clarify with Legal/Finance
R1b denominator (claim_amount vs gross_open_AR)	Reconcile rule text with implementation
ASR hard-floor discard threshold	Not numerically specified in source

Physical courier chain automation	Manual steps remain; system blocks ineligible pre-courier
-----------------------------------	---

— End of Document —

ARC Business Requirements Document v2.0 | 28-Aug-2026